

Capability-Conditioned Scaling-Down Laws for LLM Compression

Anonymous authors

Paper under double-blind review

Abstract

Capability-specialized agents rarely need frontier-scale generality: a coding or retrieval agent needs a specific capability profile at the lowest deployment cost, usually obtained by compressing a dense source model through pruning, quantization, or distillation with recovery training. Choosing among these methods is today an expensive grid search, because compression changes capabilities anisotropically and existing scaling laws predict only aggregate loss within a single method. We systematically investigate whether the model state observable before compression, together with the intervention configuration, can predict the capability-conditioned loss after pruning, quantization, and distillation, testing source-state transfer and unseen-configuration prediction separately, with the per-capability conditional loss vector as the common prediction endpoint. On a controlled panel, a source-conditioned pruning predictor predicts unseen compression densities and clearly outperforms strength-only baselines, while a same-input comparison shows no additional advantage for a shared power form. The predictive gain of the quantization candidate depends on the precision regime, and damage-magnitude transfer under aggressive quantization remains unstable. The distillation reuse-count coordinate improves QA-loss prediction on an unseen data pool but does not beat a constant baseline for math and code and carries a budget-dependent bias. Across twelve checkpoints from four families (270M–32B), method-agnostic laws on storage and active-parameter ratios fail to extrapolate across the methods tested. These results delimit the information the current predictors need, the range over which they transfer, and their limits, providing tested candidate relations and an evaluation basis for developing capability-conditioned scaling-down laws.

1 Introduction

Deployed LLM-based agents are specialists. A coding agent depends on code synthesis, debugging, and tool use; a retrieval assistant depends on multi-hop question answering; neither needs frontier-scale generality, and both are billed by inference cost. The practical question of the small-model era is therefore not “how good can an N -parameter model be?” but “what is the cheapest model whose capability profile covers this workload?” The standard route to such a model starts from a strong dense source and scales it down: pruning, quantization, or distillation, usually followed by recovery training.

Choosing among these routes is currently guesswork or grid search. Compression does not shrink all capabilities uniformly: at equal size, memory, or theoretical compute, different methods preserve different capabilities, improve some while collapsing others, hit cliffs at different strengths, and recover differently under continued training (§5). Aggregate perplexity, average benchmark scores, and effective parameter counts do not reveal whether the capabilities an agent needs survive. The full search space (source model $\times$ method $\times$ strength $\times$ recipe $\times$ recovery budget) must today be explored by running every configuration through capability benchmarks and agent evaluations, which is expensive and still prone to missing sign flips, non-monotonicities, and collapse cliffs.

This paper develops predictive scaling-down laws to replace that search. The target map is

$$\underbrace{(\text{source state})}_{\mathbf{x}_0}, \underbrace{(\text{common coordinates})}_{\mathbf{z}_m}, \underbrace{(\text{method controls})}_{\theta_m} \mapsto \underbrace{\text{capability vector}}_{\hat{\mathbf{L}}^{(m)}}, \quad (1)$$

followed by a workload link $S_W = G_W(\hat{\mathbf{L}}, \mathbf{r}_W)$ and the deployment decision $\arg \min_{m, \theta_m} C_m(\theta_m; H)$ subject to $G_W(\hat{\mathbf{L}}^{(m)}, \mathbf{r}_W) \geq S_W^{\min}$, where H is the hardware environment, $\mathbf{r}_W$ the workload’s capability requirement, and C_m the deployment-plus-recovery cost. Three design rules keep the laws usable. First, prediction-time inputs must be observable before compression: public source descriptors and an observed dense capability anchor $\mathbf{L}_0^{\text{obs}}$ that absorbs unknown data quality, optimization, and post-training history, so predictions take the form $L_{c,0}^{\text{obs}} + \widehat{\Delta L}_c^{(m)}$. Gradient and Fisher quantities serve as mechanism evidence, not as prediction inputs. Second, the observable is a vector of continuous capability-conditional losses (§3), because a decade of downstream-predictability results shows accuracy-space fitting destroys the regularity laws need (Schaeffer et al., 2025; Gadre et al., 2025; Bhagia et al., 2025). Third, cross-method unification is a hypothesis to test, not an assumption: we compare a fully shared law, a hierarchical shared-plus-corrections law, and method-specific laws by held-out prediction error (§4.2), and if unification fails, the method-selection map is the central deliverable rather than an appendix.

What the evidence shows so far. Across twelve dense checkpoints from four model families (Qwen3, Gemma-3/4, Muse, OLMo-3; 270M–32B):

1. Capability responses to compression are anisotropic with an invariant ordering. Multi-hop QA is the most pruning-robust capability and math/code the most fragile in every model measured, while magnitudes and signs are family-specific: QA in Qwen3 and OLMo-3-32B improves under substantial pruning, whereas QA in Gemma never does. Sign, cliff location, and family identity all matter to method choice.
2. Capability-conditioned prediction targets real structure. Independent benchmarks of the same skill share gradient signatures that anti-align across skills (seven models, three families); in a first single-model test, zeroing one capability’s exclusive 0.05% of coordinates selectively removed that capability (near-diagonal ablation matrix). Multi-model control ablations are in progress before stronger causal language is warranted.
3. The collapse cliff is real, proportional, and non-perturbative in loss, yet a geometric retention statistic decays smoothly and crosses a family-consistent critical value at the cliff, which suggests that the cliff boundary can be carried by one constant per family rather than fitted per model. (preliminary: three models, two families so far)
4. The other arms echo the structure. Quantization is capability-selective in the same directions with a collapse at 3 bits in all nine models measured; distillation recipes that strip chains of thought selectively damage math (paired early runs); recovery training follows a saturating law whose asymptotic residual under a fixed recipe separates redeemable from persistent damage, and capability-aligned recovery data heals faster at small budgets but re-damages under over-training.

Central thesis. The per-capability conditional loss vector serves as the common prediction endpoint for all three arms, and the arms share one information budget and one validation standard; the response form and its range of validity are then determined separately for each arm by evidence. Laws using only resource ratios fail to extrapolate across the methods we tested, which negates the tested resource-ratio models but leaves open whether some other single form unifies the arms.

Contributions. (1) A capability-conditioned scaling-down formulation: predict a multidimensional post-compression capability profile from observable source state, common compression coordinates, and method-specific controls, anchored at the observed dense capability (§3). (2) Observable-input, method-specific predictive laws for pruning, quantization, and distillation (plus recovery), each reported by held-out prediction error rather than in-sample fit (§4,§5), including a controlled two-axis test over source state and intervention strength with frozen predictions and same-budget baselines (§5.5). (3) A demonstration of cross-method non-equivalence on tested storage/active coordinates alongside

a shared capability interface (the per-capability loss endpoint), adjudicated by leave-one-method/family/size-out tests. (4) A discussion of how these predictors would enter a capability- and cost-aware method-selection map, with the existing Pareto and mis-ranking analyses as implications of prediction error rather than a validated selector. Dense-reference scaling and Fisher-implementation optimization are not claimed as contributions. Items not completed in this round (the cross-method M1/M2/M3 comparison and agent-workload selection validation) are stated as limitations in §5.6, not assumed.

2 Related Work

Scaling laws for compression, in loss space. Established compression laws are laws of pre-training loss: sparsity acts as an effective-parameter multiplier (Frantar et al., 2024), precision as a saturating effective-parameter count with a post-training degradation term (Kumar et al., 2025; Dettmers & Zettlemoyer, 2023), and sparse/quantized formats unify under a single capacity scalar (Panferov et al., 2025); distillation obeys a broken power law in teacher loss (Busbridge et al., 2025); pruned models admit post-training recovery laws (Chen et al., 2025), and pruning plus distillation-based recovery is highly effective in practice (Muralidharan et al., 2024; Xia et al., 2024). None of these validate their forms on downstream capabilities. Empirical claims of “compression laws” with downstream metrics exist (Sengupta et al., 2025) but report no held-out predictive validation and no capability stratification; task-stratified PTQ laws (Zhou et al., 2026) are the closest antecedent, limited to quantization without a mechanism or recovery axis. Our framework is complementary: it derives per-capability forms from measured geometry and validates them by extrapolation.

Predictability of downstream capabilities. Downstream accuracy is predictable essentially only through continuous log-likelihood measurements: emergent jumps are largely metric artifacts (Schaeffer et al., 2023), accuracy-space prediction degrades through a chain of information-destroying transforms (Schaeffer et al., 2025), and successful pipelines fit loss-space power laws composed with sigmoid/exponential links (Gadre et al., 2025; Bhagia et al., 2025), aggregate many tasks (Owen, 2024), or use multi-model observational latents (Ruan et al., 2024). Even then, linear loss-to-downstream relationships hold in only a minority of settings (Lourie et al., 2025). Our measurement layer adopts these lessons wholesale (§3).

Capability-selective effects of compression. Pruning disproportionately harms knowledge-intensive tasks (Jaiswal et al., 2024), low-bit quantization disproportionately harms reasoning (Liu et al., 2025), small-magnitude weights encode hard-task knowledge whose removal is not recovered by further training (Yin et al., 2024), and selective low-rank truncation can improve specific behaviors (Sharma et al., 2024). These observations motivate a capability-vector law and anticipate our signed first-order term; what has been missing is a quantitative, predictive account, which capability geometry supplies.

Curvature, gradients, and capability structure. Second-order weight importance underlies classical and modern pruning (LeCun et al., 1989; Hassibi & Stork, 1992; Frantar & Alistarh, 2023; Sun et al., 2024), and Fisher-aligned subspaces improve compression choices (Shihab et al., 2026); per-task Fisher guidance exists in vision transformers (Zhang et al., 2026). Gradient structure also localizes skills: gradients occupy low-dimensional subspaces (Gur-Ari et al., 2018; Zhao et al., 2024), gradient clustering discovers skill quanta (Michaud et al., 2023), and task vectors localize capabilities in weight space (Ilharco et al., 2023). To our knowledge, no prior work connects capability-conditional gradient/Fisher structure of the source model to cross-method compression damage and recovery; this is the gap the present paper occupies.

Origins of scaling-law forms. We follow the tradition that credible scaling-law forms carry mechanism: error decompositions (Hoffmann et al., 2022), solvable models and data-spectrum arguments (Sharma & Kaplan, 2020; Bahri et al., 2024; Maloney et al., 2022), discrete-skill models (Michaud et al., 2023), and axiomatic limit constraints (Busbridge et al., 2025); surveys document the fragility of purely fitted forms (Li et al., 2025; Besiroglu

et al., 2024). Our two-regime law with geometry-constrained coefficients is designed to meet this bar.

3 Problem Formulation

3.1 Capability vector and observables

We fix a capability set $\mathcal{C}$ (here mathematical reasoning, code generation, multi-hop QA) and give each capability c a probe set $\mathcal{P}_c$ of prompt-reference pairs from held-out benchmark splits, cross-benchmark disjoint from all transfer and recovery data (math: GSM8K/MATH-500/SVAMP; QA: HotpotQA/2Wiki/TriviaQA; code: HumanEval/MBPP, with a third independent benchmark planned). The primary observable is the capability-conditional task loss

$$L_c(w) = \frac{1}{T_c} \sum_{(x,y) \in \mathcal{P}_c} \sum_{t=1}^{|y|} -\log p_w(y_t \mid x, y_{<t}), \quad (2)$$

continuous by construction (Schaeffer et al., 2025; Gadre et al., 2025). Within-model compression deltas ΔL_c compare a model to itself and are tokenizer-consistent; any cross-family comparison of absolute levels uses tokenizer-normalized units (bits or nats per byte). Task success enters only through a separately validated monotone link $A_c = g_c(L_c)$ (not completed in this round: held-out validation of g_c on compression configurations). Probe sets are split once into estimation and measurement halves so mechanism statistics and evaluations never share sampling noise.

3.2 Three layers of prediction inputs

All prediction-time inputs must be available before compression, cheap to obtain, well defined, and public for open models.

1. Source-model descriptors. $\mathbf{x}_0 = (N_0, D_0, f, a, \mathbf{L}_0^{\text{obs}})$: non-embedding parameter count, pretraining tokens where disclosed, family, a few architecture descriptors, and the observed dense capability anchor $\mathbf{L}_0^{\text{obs}}$, the capability vector measured once on the dense model. The anchor absorbs data quality, optimization, and post-training history that metadata cannot capture; the practical prediction setting is

$$\hat{L}_c^{(m)} = L_{c,0}^{\text{obs}} + \widehat{\Delta L}_c^{(m)}(\mathbf{x}_0, \mathbf{z}_m, \boldsymbol{\theta}_m). \quad (3)$$

Descriptors with insufficient independent variation in the grid (D_0 is largely confounded with f in public families) are kept only if they improve held-out prediction. (not completed in this round: dense-reference comparison D1–D4 and the (N, D) -identification grid)

2. Common compression coordinates. $\mathbf{z}_m = (r_{\text{active}}, r_{\text{storage}}, r_{\text{compute}}, D_A)$: active-parameter ratio, storage ratio, theoretical inference-compute ratio, and post-compression adaptation tokens, all computable from the configuration alone, and the coordinates on which cross-method comparison and any shared law must live.

3. Method-specific controls. $\boldsymbol{\theta}_m$: pruning (density d ; pattern and criterion fixed to global magnitude as the canonical protocol), quantization (bit-width b ; per-output-channel symmetric RTN, weights only), distillation (student size N_S , trace tokens D_S , teacher capability vector $\mathbf{L}^T$, supervision recipe ρ), recovery (tokens D_R , data source ρ_R , fixed LoRA update recipe). Algorithmic choices that are not axes of the law are frozen, not fitted.

Gradient, Fisher, deleted-mass, and alignment quantities appear in this paper only as mechanism analysis (§4.4) and as upper-bound comparators for the reduced-form laws; they are not prediction-time inputs.

3.3 Workloads and the selection problem

A workload W specifies a requirement vector $\mathbf{r}_W$ over $\mathcal{C}$ and a link $S_W = G_W(\mathbf{L}, \mathbf{r}_W)$ from capability losses to agent success, calibrated on a subset of workloads and tested on held-out tasks and configurations (not completed in this round: agent-level validation). The decision problem the laws serve is

$$\arg \min_{m, \theta_m} C_m(\theta_m; H) \quad \text{s.t.} \quad G_W(\hat{\mathbf{L}}^{(m)}, \mathbf{r}_W) \geq S_W^{\min}, \quad (4)$$

with C_m covering storage, inference compute, measured latency where supported, and recovery training cost; unstructured-pruning parameter reductions are not assumed to yield real latency gains.

4 Predictive Scaling-Down Framework

4.1 Dense reference candidates

The dense reference is a capability-conditioned specialization and validation of established scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022; Gadre et al., 2025), not a contribution of this paper. We compare, by held-out prediction error: D1, size-only $E_c + A_c N^{-\alpha_c}$; D2, Chinchilla-style $E_c + A_c N^{-\alpha_c} + B_c D^{-\beta_c}$; D3, family-conditioned D2 with offsets $u_{c,f}$; and D4, the observed dense capability anchor $\mathbf{L}_0^{\text{obs}}$ used directly per equation 3. D2/D3 are claimed only where N , D , and f vary independently enough to identify them; D -family collinearity is reported rather than hidden. (not completed in this round: D1–D4 comparison; controlled (N, D) grid) The practical setting in this paper is D4: anchored prediction of compression-induced change.

4.2 Shared-plus-method-specific law hierarchy

We do not presume that methods unify. The general form is

$$\hat{\mathbf{L}}^{(m)} = \mathbf{L}_0^{\text{obs}} + F_{\text{shared}}(\mathbf{x}_0, \mathbf{z}_m) + F_m(\mathbf{x}_0, \mathbf{z}_m, \theta_m), \quad (5)$$

and three hypotheses are compared by held-out prediction error (never by in-sample R^2): M1, fully shared: after transformation to common coordinates $\mathbf{z}_m$, one response function covers all methods; M2, hierarchical: shared macro-structure with capability, family, and method corrections; M3, method-specific functional forms unified only by the capability-vector output, source interface, validation protocol, uncertainty reporting, cost model, and agent-performance link. (not completed in this round: M1/M2/M3 held-out comparison, including leave-one-method-out and matched-storage / matched-active-parameter / matched-initial-damage tests) If unification fails at M1/M2, the M3 laws are reported separately, and their use in a method-selection map is left to discussion (§4.6). The cross-method tests completed in this paper cover the resource-ratio models at matched storage and active-parameter coordinates; the full M1/M2/M3 held-out comparison is a stated limitation (§5.6).

4.3 Candidate functional forms per arm

Pruning (two regimes). Pre-cliff candidates, fit on the positive-damage branch and compared by the held-out battery (fit-shallow-predict-deep, leave-largest-model-out, leave-one-family-out, sign prediction, cliff-density prediction; baselines: raw sparsity, remaining parameters, removed weight norm, anchor-only, size-plus-density, family-conditioned reduced form): simple reduced forms in $(d; N_0, f)$, and, as mechanism comparators only, the measured-alignment-plus-deleted-mass form $\Delta L_c = a_c(d) + B m_c(d)^\gamma$ whose in-sample fits motivated this study (§B.2). Cliff modeling compares a hard threshold against a smooth sigmoid in d and, following the geometric order-parameter evidence (§B.4), a family-level critical-retention parameterization of $d_c^*(N_0, f)$. (not completed in this round: held-out battery results)

Quantization. Candidates at bit-width b : $q_c 4^{-b}$, a learned exponential $q_c e^{-kb}$, polynomial degradation, and an effective-parameter multiplier; validated by leave-one-bit-out and held-out model/family, with a matched-storage comparison against pruning. (not completed in this round: functional-form comparison)

Distillation. Per capability, a template with student capacity floor, teacher capability vector $\mathbf{L}^T$, a data power law in trace tokens, and a supervision-dependent coverage factor $\chi_c(\rho)$ computed from data composition (never inferred post hoc from results); capacity-gap corrections follow Busbridge et al. (2025). Confounds (total tokens, update count, answer format, teacher identity) are controlled by token-matched and update-matched runs. (not completed in this round: full capacity $\times$ teacher $\times$ data $\times$ recipe grid)

Recovery. The saturating transform, anchored at $D_R=0$:

$$\Delta \hat{L}_c^{(m)}(D_R) = \Delta L_c^{(m)}(0) \left[r_{m,c} + (1 - r_{m,c})(1 + D_R/D_{0,m,c})^{-\beta_{m,c}} \right], \quad (6)$$

where $r_{m,c}$ is the asymptotic residual fraction under the fixed recovery recipe, not a claim of fundamentally unrecoverable capability. Recovery-source dependence (general vs. capability-aligned vs. mixed data) is an explicit axis.

Uncertainty. Configuration-level cluster bootstrap for fitted coefficients; benchmark bootstrap for deterministic compression; multiple seeds for training-based arms at small scale; calibration coverage reported alongside MAE and relative error.

4.4 Mechanism analysis: capability geometry

As supporting evidence rather than a prediction interface, we measure the capability-conditional mean gradient $\bar{g}_c$ and Fisher structure $F_c = \mathbb{E}_{i \sim c}[g_i g_i^\top]$ at the dense checkpoint, streamable at 30B scale (diagonal spectra in magnitude-quantile bins; quadratic forms via $\delta w^\top F_c \delta w = \mathbb{E}_i[(g_i^\top \delta w)^2]$). Three uses: (i) testing whether capabilities occupy distinct, partially separable parameter regions (§B.1); (ii) the signed first-order term $\bar{g}_c^\top \delta w$, which no positive-semidefinite effective-parameter model can express and which anticipates capability-selective sign effects (§B.3); and (iii) upper-bound comparators for reduced-form laws: if a reduced form approaches the accuracy of the mechanism form without gradient access, the practical interface loses nothing.

4.5 Fitting and validation protocol

Functional forms and keep/reject rules are committed before fits; model comparison uses repeated held-out extrapolation (largest size, deepest compression, largest recovery budget, leave-one-family-out), never in-sample fit indices (Li et al., 2025; Besiroglu et al., 2024); manuscript counts of models, benchmarks, and fits are generated from result files, not typed by hand; negative results are reported, not tuned away.

4.6 Discussion: toward a method-selection map

Intersecting per-method predictors at matched cost coordinates would partition the (compression budget $\times$ capability requirement) space into regions where pruning, quantization, distillation, or a composition is cheapest, with uncertainty bands, and would be evaluated against exhaustive search by constraint-satisfaction rate, cost regret, selected-method accuracy, and the number of compression experiments avoided. This is how the predictors of this paper would be used, not a result of it: no held-out agent or method-selection validation was run (§5.6), and the Pareto and mis-ranking analyses we report are implications of prediction error rather than a validated selector.

5 Experiments

The experiments follow the prediction problem of §3. We first describe the panel and protocol (§5.1), then the capability-conditioned responses of the heterogeneous panel under

Model	Family	ΔL_c at $d = 0.7$			$\Delta L_m @ 0.5$	d_{math}^*	QA sign, mild d
		math	code	QA			
Qwen3-0.6B	Qwen3	+0.49	+0.56	-0.26	+4.7	0.60	improves
Qwen3-1.7B	Qwen3	+0.24	+0.19	-0.51	+3.6	0.60	improves
Qwen3-4B	Qwen3	-0.00	-0.09	-0.79	+1.3	0.50	improves
Gemma3-270M	Gemma-3	+3.49	+4.04	+2.54	+11.2	0.70	degrades
Gemma3-1B	Gemma-3	+1.31	+1.23	+0.83	+13.8	0.70	degrades
Gemma3-4B	Gemma-3	+0.83	+1.47	+0.24	+7.0	0.60	degrades
Gemma3-12B	Gemma-3	+0.68	+0.46	+0.54	+17.1	0.60	degrades
Gemma3-27B	Gemma-3	+4.04	+4.69	+3.59	+15.2	0.70	degrades
Gemma4-31B	Gemma-4	+0.68	+0.77	+2.89	+16.1	0.60	degrades
Muse-30B	Muse	+0.17	+0.14	+0.20	+3.5	0.55	degrades
OLMo3-7B	OLMo-3	+0.03	+0.03	+0.14	+0.22	0.35	degrades
OLMo3-32B	OLMo-3	+0.02	+0.03	+0.02	+0.09	<0.30	improves (deep)

Table 1: Per-capability pruning damage (nats/token). Three invariances hold everywhere: QA is the most robust capability pre-cliff; every model meets a cliff in a narrow proportional band (OLMo3-32B has not met one by $d = 0.3$, where its QA improves by -0.24); and magnitudes and signs are family coefficients.

pruning (§5.2), quantization (§5.3), and distillation (§5.4), and then the controlled two-axis prediction tests that carry the main claims (§5.5), ending with the limitations that remain (§5.6). Mechanism evidence (capability regions and selective ablation, the gradient-alignment law, the signed first-order term, the geometric cliff order parameter), the recovery arm, scale and family structure, and the re-analysis of the prior ability-space grid are supporting material and are given in full in Appendix B.

5.1 Setup

Models. Twelve dense checkpoints from four families: Qwen3 {0.6B, 1.7B, 4B} (measured before a cluster-policy change restricted further use; retained as a comparison family), Gemma-3 {270M, 1B, 4B, 12B, 27B}, Gemma-4-31B, Muse-Glimmer-30B, OLMo-3 {7B, 32B}. **Probes.** Three capabilities (math, code, multi-hop QA), each probed by held-out splits of benchmarks cross-benchmark disjoint from all transfer data; for capability-region tests, independent benchmarks per capability (math: GSM8K/MATH-500/SVAMP; QA: HotpotQA/2Wiki/TriviaQA; code: HumanEval/MBPP, a third planned) plus a C4 control. **Compression.** Global magnitude pruning (ten densities, $d \in [0.3, 0.9]$); per-output-channel round-to-nearest weight quantization ($b \in \{8, 6, 4, 3\}$); black-box distillation from two frontier API teachers onto Gemma-3 students; LoRA recovery training on general (C4) or teacher-trace data. **Geometry.** Estimated as in §4.4; estimation and measurement probe halves are disjoint throughout, and a checkpoint-integrity guard aborts any run with missing or re-initialized weights.

5.2 Pruning damage curves

Table 1 reports per-capability damage at reference densities and the math cliff onset d^* (largest density with $\Delta L_{\text{math}} \geq 1$).

5.3 Quantization arm

Across nine models measured at $b \in \{8, 6, 4, 3\}$: int8 and int6 are within noise of dense everywhere; int4 damages small models (Gemma3-270M math +2.3) and barely touches large ones (Gemma3-27B math +0.33); and $b = 3$ collapses every model measured (+5 to +18 nats). In leave-one-bit-out validation a learned exponential $q_c e^{-kb}$ beats the fixed $q_c 4^{-b}$ candidate (pooled MAE 6.73 vs. 7.63): bit decay is exponential, with a fitted base. Capability selectivity transfers to this basis: QA improves under int4 for Qwen3-1.7B (-0.21) and Gemma3-27B (-0.67), both clearing the mildest-perturbation noise screen (29 of 62

nominal improvement cells across all arms survive that screen; the rest are within noise and not claimed).

5.4 Distillation arm

Black-box traces from two frontier teachers (GPT-5.6-luna and Claude-Sonnet-4-6; 600 per domain each; single-teacher runs in a paired design) train Gemma-3 students under coverage recipes. The paired results are directionally consistent with the pre-registered coverage-deletion prediction: the answer-only recipe hurts math more than full-trace training (+0.19 to +0.22 vs. +0.09 to +0.12), while QA improves in every configuration (−0.58 to −1.44). We do not attribute that gap to reasoning-coverage loss on this evidence alone, because the recipes differ in supervised tokens and update counts; the controlled comparisons that hold those quantities fixed are the reuse-count and pool experiments of §5.5, where the response is modeled as a signed transfer relative to the student’s own dense loss.

5.5 Controlled prediction on two axes

The heterogeneous panel above tests how far a law transfers across finished models; it cannot separate the effect of training history from that of architecture and data, because those covary across public families. We therefore add a controlled panel, Pythia {160M, 410M, 1.4B} at pretraining steps {16k, 64k, 143k} (33.6B, 134B, and 300B tokens), on which the source state $\mathbf{x}_0 = (N_0, D_0, L_{c,0})$ varies along known axes, and we ask two questions separately: whether source-state information predicts the compression response at a fixed intervention (the source axis), and whether a relation fit at seen intervention strengths predicts unseen ones (the strength axis). Table 2 collects every registered test. Predictions marked P were frozen before the corresponding measurement, L are leave-one-out on an existing panel, and R are baselines added at closeout with the original prediction untouched. All three arms use the same K0 budget (pre-compression information only), the same endpoint, and the same rule: a candidate is credited only against the strongest baseline at the same information budget, that is, one built from the same inputs.

Source axis. Three statements are kept separate here: whether adding D_0 improves on the model without it, whether the full-input model improves on the strongest simple baseline, and whether a frozen model succeeds on a new source state. On the 3×3 panel, adding D_0 to $\{N_0, L_{c,0}\}$ lowers held-out error in 4 of 12 arm $\times$ capability $\times$ split comparisons with a bootstrap interval excluding zero, for math and code but not QA. Against the strongest simple baseline, a per-configuration median, the full predictor wins with an interval excluding zero only for code; for math the point improvement is positive but the interval includes zero. The gain concentrates in aggressive settings (pruning $d = 0.6$, int3), and for the reported capabilities and splits it is absent when the most-trained checkpoint is held out; we do not generalize this to all latest-stage extrapolation. Because the predictor uses configuration indicators, it is a source-transfer model at a fixed density or bit-width, not a law over the intervention axis. On a never-fit training stage (step 96k, a frozen prospective stage interpolation between 64k and 143k), the frozen pruning predictor beat both the baseline and the no- D_0 model for all three capabilities (math 0.090 vs. 0.367; code 0.128 vs. 0.438; QA 0.426 vs. 0.832), whereas the quantization predictor was mixed (math and QA above baseline, code below), a pattern the partition analysis below explains.

Pruning: strength axis. A shared-shape candidate $\widehat{\Delta L}_c(\mathbf{x}, d) = A_c(\mathbf{x}) ((1 - d)/0.3)^{\gamma_c}$ with source-conditioned amplitude $A_c(\mathbf{x}) = \beta_c \cdot (1, \log N_0, L_{c,0}, \log D_0)$ was frozen on the panel at $d \in \{0.9, 0.8, 0.7, 0.6\}$ and evaluated at unseen $d = 0.65$ (interpolation) and $d = 0.55$ (deeper extrapolation), including a new source (410M at step 96k). It beats every strength-only baseline by a wide margin (MAE 0.371, 0.524, and 0.759 for math, code, and QA against 1.37, 1.63, and 1.61 for a strength-only curve), because a source-blind curve cannot see that one state (160M at step 143k) is several times more fragile than another. Two closeout baselines at the same information budget, however, change what this shows. A per-density source regression with linear interpolation and extrapolation in d , which uses the same inputs and no shared exponent, has better point estimates than the candidate

Method	Cap.	Test	Split	Cand.	Strongest budget baseline	same-Base.	Impr. St.
prune	math	source transfer, fixed d (3×3, source/interp leave-one-step-out)		0.253	per-config median	0.413	+0.160 L
prune	code	source transfer, fixed d (3×3, source/interp leave-one-step-out)		0.292	per-config median	0.516	+0.224 L
prune	qa	source transfer, fixed d (3×3, source/interp leave-one-step-out)		0.806	per-config zero	0.500	-0.306 L
prune	math	unseen d 0.65/0.55, incl. new strength+source/int.+ext. source		0.371	A2: per- d regr. + lin. interp	0.283	-0.088 P
prune	code	unseen d 0.65/0.55, incl. new strength+source/int.+ext. source		0.524	A2: per- d regr. + lin. interp	0.279	-0.244 P
prune	qa	unseen d 0.65/0.55, incl. new strength+source/int.+ext. source		0.759	A2: per- d regr. + lin. interp	0.818	+0.059 P
quant	math	source transfer, fixed bit, ≥ 4 -bit (3 new sources)	source/interp	0.029	per-bit median	0.084	+0.055 P
quant	math	source transfer, fixed bit, int3 (3 new sources)	source/interp	1.488	per-bit median	1.934	+0.446 P
quant	code	source transfer, fixed bit, ≥ 4 -bit (3 new sources)	source/interp	0.104	per-bit median	0.094	-0.010 P
quant	code	source transfer, fixed bit, int3 (3 new sources)	source/interp	1.963	per-bit median	0.843	-1.120 P
quant	qa	source transfer, fixed bit, ≥ 4 -bit (3 new sources)	source/interp	0.173	per-bit median	0.134	-0.039 P
quant	qa	source transfer, fixed bit, int3 (3 new sources)	source/interp	2.046	per-bit median	1.727	-0.319 P
distill	math	δ source transfer (3×3 LoRA, source/interp leave-one-step-out)		0.014	constant (median)	0.031	+0.017 L
distill	code	δ source transfer (3×3 LoRA, source/interp leave-one-step-out)		0.057	constant (median)	0.061	+0.004 L
distill	qa	δ source transfer (3×3 LoRA, source/interp leave-one-step-out)		0.263	constant (mean)	0.159	-0.104 L
distill	math	unseen pool U225 from config(pool)/interp U75/U600		0.122	E-only	0.191	+0.069 P/R
distill	code	unseen pool U225 from config(pool)/interp U75/U600		0.069	E-only	0.117	+0.048 P/R
distill	qa	unseen pool U225 from config(pool)/interp U75/U600		0.706	constant	1.449	+0.743 P/R
prune	all	NEW source pythia-1b@96k, source+strength/interp d=0.65 (interp)		0.193	strongest: (gamma=1) input	A1 0.149	-0.044 P-new
prune	all	NEW source pythia-1b@96k, source+strength/extrap d=0.55 (extrap)		1.302	strongest: strength-only (same-input best A2 1.139)	0.273	-1.029 P-new
quant	all	NEW source pythia-1b@96k, source/interp int4 (seen bit)		0.074	strongest: per-bit median	0.111	+0.037 P-new
quant	all	NEW source pythia-1b@96k, source/interp int3 (seen bit)		0.397	strongest: per-bit median	0.620	+0.223 P-new

Table 2: Main prediction table (capability loss, nats/token); candidate forms and inputs (all K0) are given in §5.5. Impr. = baseline MAE – candidate MAE (positive = candidate better) against the strongest baseline at the same information budget (K0). Three statements are kept separate: whether adding D_0 improves on the model without it, whether the full-input model improves on the strongest same-budget baseline, and whether a frozen model succeeds on a new source state; Impr. reports point-estimate error reduction, and significance is stated in the text. Status: P = frozen prospective (prediction registered before measurement); L = leave-one-out on an existing panel (retrospective); R = baseline/diagnostic added at closeout with the original prediction untouched; P/R = the four candidates were frozen (P) but the per-capability choice among them was made after the test (R). Every number is generated from results/v*/summary.json by analysis/v45_main_table.py; per-source, per-density, and per-pool breakdowns are in the corresponding docs. Point estimates; the controlled panels have three size/step/pool clusters, so intervals are panel-conditional (see text).

for math (0.283 vs. 0.371) and code (0.279 vs. 0.524) and is within noise for QA (0.818 vs. 0.759); whether these differences are significant is not established on a panel of three sources. The candidate beats only the variant with γ_c fixed to one. On the tested panel, therefore, the source-informed candidate outperforms strength-only baselines, and the added same-input comparison shows no additional advantage for the shared power form. This does not mean that only the inputs matter, since the per-density baseline also uses the response structure across the measured strengths. At $d = 0.55$ the signed errors are mixed (six under-predictions, three over-predictions), with the most fragile source over-predicted, so we do not describe the deep regime as systematically under-predicted. QA amplitude does not follow the math/code relation.

Quantization: partition. The tested interventions are the integer bit-widths $\{8, 6, 4, 3\}$; there is no integer point between 3 and 4, so the strength-axis test available here is per configuration. On three new sources at step 96k, responses at 4 bits and above are near zero and candidate and per-bit-median errors are both small (math 0.029 vs. 0.084; code 0.104 vs. 0.094; QA 0.173 vs. 0.134), so no incremental value over a simple baseline is demonstrated at that error scale, which is not evidence that none exists. The int3 responses are real collapse, and the candidate advantage there is unstable across sources (+0.45 for math, -1.12 for code, and -0.32 for QA relative to the median), so magnitude transfer at int3 is not established. The fixed 4^{-b} shape misfits under this quantizer definition, and a learned exponential fits better within the tested range (§5.3). We claim no collapse-risk law.

New-source frozen prospective (pruning and quantization). To test the retrospective same-input comparison on an independent source, we froze and committed every candidate’s prediction for Pythia-1B at step 96k, a new size inside the 160M–1.4B range at an interpolated training stage that had never entered any fit or selection, before measuring it, using only its measured dense loss and public descriptors. At the interpolated density $d = 0.65$ the source-conditioned forms succeed and are indistinguishable (MAE 0.149 for the $\gamma = 1$ variant, 0.172 for the per-density baseline, and 0.193 for the shared power form, against 0.704 for strength-only and 0.657 for zero change), with math and code errors of 0.02 to 0.09 and QA the weakest capability. At the extrapolated density $d = 0.55$ every source-conditioned candidate fails (1.14 to 1.69), under-predicting the collapse of this source by 1 to 1.5 nats and missing the sign of the QA response, while the source-blind strength-only curve is closest (0.273); the transfer range of the source-conditioned family is therefore the interpolation regime. For quantization the frozen full-input predictor is best at both bits on this source (int4 MAE 0.074 against 0.211 without D_0 and 0.111 for the per-bit median; int3 0.397 against 0.620 and 2.46), so D_0 carries information here. This is one source-state: it confirms or refutes transfer for that state only and supports no population claim.

Distillation: source and pool axes. With the reference fixed to the initial student, $\delta_c = L_c(S_{KD}) - L_c(S_0)$, a controlled LoRA panel (the same 3×3 sources with fixed teacher, pool, and reuse count $E = T/D_U$, the ratio of processed tokens to pool tokens) admits a source-transfer predictor only for math, where a linear model in $\{N_0, L_{c,0}, D_0\}$ halves the held-out error of a constant (0.014 vs. 0.031); for code the two are within noise (0.057 vs. 0.061) and for QA a per-capability constant is better (0.159 vs. 0.263). On the pool axis, predictors frozen on the endpoint pools (U75, U600) were tested on an unseen middle pool (U225). Its reuse count of 2.3–4.7 lies inside the endpoint range, so this is a configuration interpolation, and because the endpoints are pool prefixes while U225 is randomly sampled, the sampling protocol also changes. The reuse-count predictor beats a constant only for QA (0.706 vs. 1.449); for math and code a constant is best (0.122 and 0.069) and the endpoint-fit forms over-extrapolate. All four frozen candidates (a constant, a budget-only form, a reuse-only form, and a two-variable form) were registered before the U225 measurement; the per-capability recommendation of reuse-only for QA and a constant for math and code is a selection made after seeing those results and is labeled as such in Table 2. The prediction residual is dominated by systematic bias rather than by pool variation: math and code show a small over-prediction bias with little spread, and the QA reuse-count predictor carries a bias that changes sign across the two budgets (-0.87 then +0.48), a form misfit rather than pool sensitivity. Around that bias, variation across pools exceeds variation across training

seeds, but with three pools and two seeds this describes the design rather than decomposing a population variance, and we make no causal claim about the residual.

Reading across arms. The three arms share the prediction endpoint, the information budget, and the validation standard, and the response form and its range of validity are determined for each arm by evidence: pruning yields a source-conditioned prediction that extends to unseen densities, quantization a per-configuration validation whose only large signal is an unstable collapse magnitude, and distillation a capability-specific transfer in which the reuse-count predictor helps QA and a constant suffices for math and code. The cross-method tests reported here cover the resource-ratio models of §4.2 at matched storage and active-parameter coordinates; the full M1/M2/M3 held-out comparison was not completed (§5.6). Whether any single response form unifies the arms therefore remains open. How these predictors would enter a capability- and cost-aware method-selection map is discussed in §4.6; the existing Pareto and mis-ranking analyses are implications of prediction error, not a validated selector.

5.6 Remaining limitations

The controlled panels have three size, step, and pool clusters, so all intervals are panel-conditional rather than population estimates. The shared pruning form is not independently necessary, since a per-density fit ties it. No held-out agent or method-selection validation was run, so the selection map is a framework deliverable rather than a validated selector. The full capacity $\times$ teacher $\times$ data distillation grid, large-model geometric order parameters, the cross-method M1/M2/M3 held-out comparison of §4.2, agent-workload selection validation, and the held-out extrapolation battery for every reduced form (§4.5) were not completed in this round and are stated as such rather than filled in.

6 Conclusion

We systematically investigated whether the model state observable before compression, together with the intervention configuration, can predict the capability-conditioned loss after pruning, quantization, and distillation, testing source-state transfer and unseen-configuration prediction separately with the per-capability conditional loss vector as the common prediction endpoint. On a controlled panel, a source-conditioned pruning predictor predicts unseen compression densities and clearly outperforms strength-only baselines, while a same-input comparison shows no additional advantage for a shared power form. The predictive gain of the quantization candidate depends on the precision regime, and damage-magnitude transfer under aggressive quantization remains unstable. The distillation reuse-count coordinate improves QA-loss prediction on an unseen data pool but does not beat a constant baseline for math and code and carries a budget-dependent bias. Across the heterogeneous panel, capability responses are anisotropic with an invariant ordering and family-specific magnitudes and signs, the capability geometry of the source model supports the measurement layer as mechanism evidence, and laws using only resource ratios fail to extrapolate across the methods tested, which negates those models while leaving open whether another single form unifies the arms. These results delimit the information the current predictors need, the range over which they transfer, and their limits, providing tested candidate relations and an evaluation basis for developing capability-conditioned scaling-down laws. How such predictors would enter a capability- and cost-aware method-selection map is left to future use (§4.6); the controlled panels are small and their intervals panel-conditional, and these limits are stated rather than filled in.

References

Yasaman Bahri, Ethan Dyer, Jared Kaplan, Jaehoon Lee, and Utkarsh Sharma. Explaining neural scaling laws. *Proceedings of the National Academy of Sciences*, 121(27):e2311878121, 2024.

- Tamay Besiroglu, Ege Erdil, Matthew Barnett, and Josh You. Chinchilla scaling: A replication attempt. arXiv preprint arXiv:2404.10102, 2024.
- Akshita Bhagia, Jiacheng Liu, Alexander Wettig, David Heineman, Oyvind Tafjord, Ananya Harsh Jha, Luca Soldaini, Noah A. Smith, Dirk Groeneveld, Pang Wei Koh, Jesse Dodge, and Hannaneh Hajishirzi. Establishing task scaling laws via compute-efficient model ladders. In Conference on Language Modeling (COLM), 2025.
- Dan Busbridge, Amitis Shidani, Floris Weers, Jason Ramapuram, Etai Littwin, and Russ Webb. Distillation scaling laws. In International Conference on Machine Learning, 2025.
- Xiaodong Chen, Yuxuan Hu, Xiaokang Zhang, Yanling Wang, Cuiping Li, Hong Chen, and Jing Zhang. P² law: Scaling law for post-training after model pruning. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025.
- Tim Dettmers and Luke Zettlemoyer. The case for 4-bit precision: k-bit inference scaling laws. In International Conference on Machine Learning, 2023.
- Elias Frantar and Dan Alistarh. Sparsegpt: Massive language models can be accurately pruned in one-shot. In International Conference on Machine Learning, 2023.
- Elias Frantar, Carlos Riquelme, Neil Houlsby, Dan Alistarh, and Utku Evci. Scaling laws for sparsely-connected foundation models. In International Conference on Learning Representations, 2024.
- Samir Yitzhak Gadre, Georgios Smyrnis, Vaishal Shankar, Suchin Gururangan, Mitchell Wortsman, Rulin Shao, Jean Mercat, Alex Fang, Jeffrey Li, Sedrick Keh, Rui Xin, Marianna Nezhurina, Igor Vasiljevic, Luca Soldaini, Jenia Jitsev, Alex Dimakis, Gabriel Ilharco, Pang Wei Koh, Shuran Song, Thomas Kollar, Yair Carmon, Achal Dave, Reinhard Heckel, Niklas Muennighoff, and Ludwig Schmidt. Language models scale reliably with over-training and on downstream tasks. In International Conference on Learning Representations, 2025.
- Guy Gur-Ari, Daniel A. Roberts, and Ethan Dyer. Gradient descent happens in a tiny subspace. arXiv preprint arXiv:1812.04754, 2018.
- Babak Hassibi and David G. Stork. Second order derivatives for network pruning: Optimal brain surgeon. In Advances in Neural Information Processing Systems, 1992.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Thomas Hennigan, Eric Noland, Katherine Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karén Simonyan, Erich Elsen, Oriol Vinyals, Jack Rae, and Laurent Sifre. An empirical analysis of compute-optimal large language model training. In Advances in Neural Information Processing Systems, 2022.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In International Conference on Learning Representations, 2023.
- Ajay Jaiswal, Zhe Gan, Xianzhi Du, Bowen Zhang, Zhangyang Wang, and Yinfei Yang. Compressing llms: The truth is rarely pure and never simple. In International Conference on Learning Representations, 2024.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. arXiv preprint arXiv:2001.08361, 2020.
- Tanishq Kumar, Zachary Ankner, Benjamin F. Spector, Blake Bordelon, Niklas Muennighoff, Mansheej Paul, Cengiz Pehlevan, Christopher Ré, and Aditi Raghunathan. Scaling laws for precision. In International Conference on Learning Representations, 2025.
- Yann LeCun, John S. Denker, and Sara A. Solla. Optimal brain damage. In Advances in Neural Information Processing Systems, 1989.

- Margaret Li, Sneha Kudugunta, and Luke Zettlemoyer. (mis)fitting: A survey of scaling laws. In International Conference on Learning Representations, 2025.
- Ruikang Liu, Yuxuan Sun, Manyi Zhang, Haoli Bai, Xianzhi Yu, Tiezheng Yu, Chun Yuan, and Lu Hou. Quantization hurts reasoning? an empirical study on quantized reasoning models. In Conference on Language Modeling (COLM), 2025.
- Nicholas Lourie, Michael Y. Hu, and Kyunghyun Cho. Scaling laws are unreliable for downstream tasks: A reality check. In Findings of the Association for Computational Linguistics: EMNLP 2025, 2025.
- Alexander Maloney, Daniel A. Roberts, and James Sully. A solvable model of neural scaling laws. arXiv preprint arXiv:2210.16859, 2022.
- Eric J. Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The quantization model of neural scaling. In Advances in Neural Information Processing Systems, 2023.
- Saurav Muralidharan, Sharath Turuvekere Sreenivas, Raviraj Joshi, Marcin Chochowski, Mostofa Patwary, Mohammad Shoeybi, Bryan Catanzaro, Jan Kautz, and Pavlo Molchanov. Compact language models via pruning and knowledge distillation. In Advances in Neural Information Processing Systems, 2024.
- David Owen. How predictable is language model benchmark performance? arXiv preprint arXiv:2401.04757, 2024.
- Andrei Panferov, Alexandra Volkova, Ionut-Vlad Modoranu, Vage Egiazarian, Mher Safaryan, and Dan Alistarh. Unified scaling laws for compressed representations. In Advances in Neural Information Processing Systems, 2025.
- Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational scaling laws and the predictability of language model performance. In Advances in Neural Information Processing Systems, 2024.
- Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In Advances in Neural Information Processing Systems, 2023.
- Rylan Schaeffer, Hailey Schoelkopf, Brando Miranda, Gabriel Mukobi, Varun Madan, Adam Ibrahim, Herbie Bradley, Stella Biderman, and Sanmi Koyejo. Why has predicting downstream capabilities of frontier ai models with scale remained elusive? In International Conference on Machine Learning, 2025.
- Ayan Sengupta, Siddhant Chaudhary, and Tanmoy Chakraborty. Compression laws for large language models. arXiv preprint arXiv:2504.04342, 2025.
- Pratyusha Sharma, Jordan T. Ash, and Dipendra Misra. The truth is in there: Improving reasoning in language models with layer-selective rank reduction. In International Conference on Learning Representations, 2024.
- Utkarsh Sharma and Jared Kaplan. A neural scaling law from the dimension of the data manifold. arXiv preprint arXiv:2004.10802, 2020.
- Ibne Farabi Shihab, Sanjeda Akter, and Anuj Sharma. Beyond variance: Knowledge-aware LLM compression via fisher-aligned subspace diagnostics. arXiv preprint arXiv:2601.07197, 2026.
- Mingjie Sun, Zhuang Liu, Anna Bair, and J. Zico Kolter. A simple and effective pruning approach for large language models. In International Conference on Learning Representations, 2024.
- Mengzhou Xia, Tianyu Gao, Zhiyuan Zeng, and Danqi Chen. Sheared llama: Accelerating language model pre-training via structured pruning. In International Conference on Learning Representations, 2024.

Lu Yin, Ajay Jaiswal, Shiwei Liu, Souvik Kundu, and Zhangyang Wang. Junk DNA hypothesis: Pruning small pre-trained weights irreversibly and monotonically impairs “difficult” downstream tasks in LLMs. In International Conference on Machine Learning, 2024.

Yipu Zhang, Jintao Cheng, Weilun Feng, Jiehao Luo, Chuanguang Yang, Zhulin An, Yongjun Xu, and Wei Zhang. Not all tasks quantize equally: Fisher-guided quantization for visual geometry transformer. arXiv preprint arXiv:2605.15828, 2026.

Jiawei Zhao, Zhenyu Zhang, Beidi Chen, Zhangyang Wang, Anima Anandkumar, and Yundong Tian. Galore: Memory-efficient llm training by gradient low-rank projection. In International Conference on Machine Learning, 2024.

Chenxi Zhou, Pengfei Cao, Jiang Li, Bohan Yu, Jinyu Ye, Jun Zhao, and Kang Liu. Task-stratified knowledge scaling laws for post-training quantized large language models. In Findings of the Association for Computational Linguistics: ACL 2026, 2026.

A Appendix

A.1 Probes and the capability-conditioned loss

Each capability is measured on a fixed probe set of prompt-reference pairs drawn once, with a fixed seed, from held-out benchmark splits that are disjoint from every training and recovery source used in the paper. The primary measurement benchmarks are MATH-500 for mathematical reasoning, MBPP for code generation, and 2WikiMultihopQA for multi-hop question answering. For each capability we draw 128 probes and split them by index parity into an estimation half, used only for gradient and Fisher statistics, and a measurement half of 64 probes on which every reported loss is computed, so that mechanism statistics and evaluations never share sampling noise. The measurement check of Appendix B adds one independent secondary benchmark per capability (SVAMP, HumanEval, and TriviaQA; 128 probes each, same seed); these were chosen to exclude the benchmarks from which the distillation traces were derived (GSM8K, CodeAlpaca, and HotpotQA), and a sample-level audit found no overlap between any measurement probe and the training pool.

The loss is the conditional cross-entropy of the reference completion given the prompt, summed over completion tokens and divided by the number of completion tokens, as in Eq. equation 2. Prompts and references are tokenized with the model’s own tokenizer and truncated to 1024 tokens; prompt tokens are masked out of the loss. All within-model responses ΔL_c are therefore differences in native-token nats on identical token sequences. Because native-token units differ across tokenizers, cross-family comparisons of absolute levels use per-byte normalization or are reported separately, and no cross-tokenizer quantity is treated as a common numerical scale. Negative responses are retained as measured; nothing is clipped at zero.

A.2 Compression protocols

Pruning. Global magnitude pruning removes the smallest weights across all transformer matrices (attention and MLP projections; embeddings, output head, biases, and norms are excluded) to a target retained density d . The global threshold is estimated from a seeded sample of absolute weights so that the same procedure scales to 30B-parameter models; every density is applied to the same dense reference weights, which are held on the CPU for the largest models so that a single 80 GB device suffices.

Quantization. Weights are fake-quantized per output channel with symmetric round-to-nearest at bit-width b , using the integer range $[-(2^{b-1} - 1), 2^{b-1} - 1]$ and a per-channel scale set by the channel’s maximum absolute value; activations are left in bfloat16. The dense weights are restored between bit-widths. Because the symmetric range uses $2^{b-1} - 1$ levels, the step-size ratio between consecutive bit-widths is $(7/15)^2 \approx 0.22$ rather than 0.25, which is why the fixed 4^{-b} shape is not the natural reference for this quantizer.

Distillation. Students are fine-tuned on black-box teacher traces (a frontier API teacher answering GSM8K, HotpotQA, and CodeAlpaca prompts; 600 traces per domain) under the full-trace recipe, with strict LoRA adapters of rank 16 and scaling 32 on the query, key, value, output, gate, up, and down projections, AdamW at learning rate 10^{-4} , an effective batch of 16 sequences, and a cosine schedule with 3% warmup. A run aborts rather than silently falling back to full fine-tuning if the adapter library is unavailable, and the actual training mode and trainable parameter count are recorded in every manifest. Token accounting distinguishes processed tokens (all input tokens seen, including repetitions), unique pool tokens, and supervised completion tokens. The data pool of U examples per domain is either the first U rows of the trace file (the historical protocol) or a random sample drawn with a pool-sampling seed that is independent of the training seed; pools are identified by the sorted hashes of their samples and pairwise overlaps are reported. The historical schedule spanned a fixed number of epochs, so its cosine horizon varied with pool size; the controlled matrix of §5.5 uses a schedule defined on the same absolute processed-token budget for every pool, and the two protocols are never mixed in one fit.

A.3 The controlled Pythia panel

The panel uses the public Pythia checkpoints at sizes 160M, 410M, and 1.4B and pretraining steps 16,000, 64,000, and 143,000, which correspond to 33.6B, 134.2B, and 299.9B processed tokens at the published batch of 1024×2048 tokens per step. The size coordinate N_0 counts transformer-matrix parameters only (the query, key, value, output, and MLP matrices of every layer), computed from the architecture configuration, and the training-tokens coordinate is $\bar{D}_0 = \text{step} \times 2,097,152$. Each checkpoint’s weight identity was verified before use: the 2.8B revisions published for different steps were found to load identical weights and were excluded, and an integrity guard now rejects any pair of nominally different checkpoints with identical measured panels. New sources added for the prospective tests (Pythia-1B at steps 32,000, 96,000, and 112,000; Pythia-6.9B at 32,000 and 112,000) were confirmed to have never entered any fit or model selection before their predictions were frozen.

A.4 Registration, freezing, and labels

Every prediction test in Table 2 follows one order. The functional forms, the development set, and every standardization and coefficient are fixed and committed first. For a new source the dense capability loss is then measured, since it is a permitted pre-compression input; the predictions for every configuration and candidate are written to a versioned file and committed to the repository, with the commit hash recorded; and only then is the compressed or distilled outcome measured. Information budgets are reported separately: K0 uses pre-compression information only, K1 adds one pre-specified target-side calibration point, and post-hoc rescaling on target outcomes is reported only as an oracle diagnostic. Rows are labeled P when the prediction was frozen before measurement in this work, L for leave-one-out on an existing panel, R for a baseline or diagnostic added after a test with the original prediction left untouched, and P/R when a set of candidates was frozen but the per-capability choice among them was made after seeing the result. When a diagnostic overturns an earlier headline, the earlier claim is downgraded in the text rather than the prediction being rewritten.

A.5 Reproduction

All tables are generated from machine-readable result files by CPU-only scripts that run in seconds and require neither model weights nor a GPU; the repository mirrors the scripts and the result files they read, and the main prediction table is produced by a single command. Model measurements are reproducible from the recorded revisions, probe seeds, and protocols above.

B Supporting analyses

B.1 Capabilities occupy distinct regions of gradient space

Convergent validity (block structure). For each of nine benchmarks we measure a gradient signature (the benchmark-conditional diagonal Fisher) and compare all pairs. After removing the component shared across benchmarks, signatures of same-capability benchmarks are positively aligned (cosine $+0.36$ to $+0.45$) while signatures of different capabilities are anti-aligned (-0.23 to -0.25), in every one of seven models across three families (Figs. 1–10); the C4 control sits between blocks. (preliminary: point estimates; example-bootstrap, permutation, and prompt-stability tests are queued (B2)) The mathematics benchmarks cluster regardless of format, and within-block fine structure matches intuition (SVAMP, grade-school word problems, is the loosest member of the math block). A projected-gradient SVD version gives the same verdict in subspace terms: principal-angle similarity within capability exceeds cross-capability similarity by $2\text{--}4\times$.

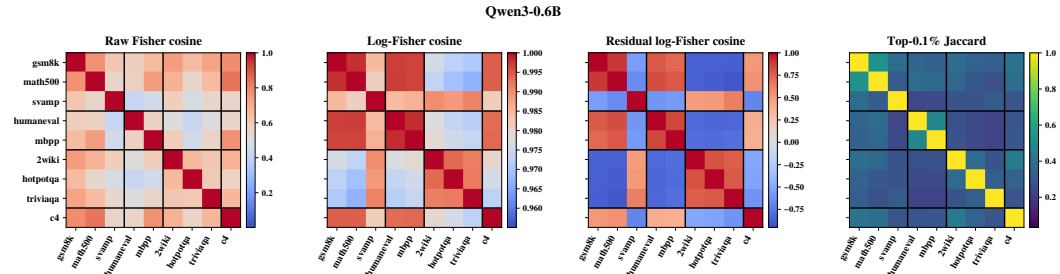


Figure 1: Capability regions in gradient space for Qwen3-0.6B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

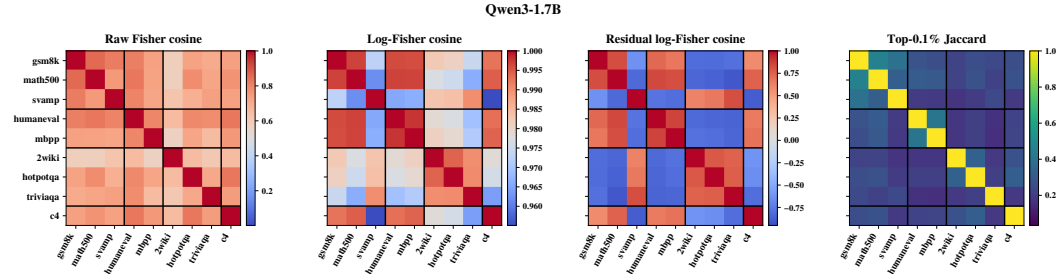


Figure 2: Capability regions in gradient space for Qwen3-1.7B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

Selective ablation (preliminary: single model; random, magnitude-matched, and global-Fisher controls pending (B1)). A first targeted test: select the top 0.05% of weight coordinates by each capability’s residual Fisher signature (excluding coordinates shared with other capabilities) and zero them. The damage matrix is nearly diagonal (Gemma-3-1B; rows ablate, columns measure, nats/token):

Ablated region	math	code	QA
math	+4.63	+0.43	+0.05
code	+0.42	+7.14	+0.00
QA	+0.01	+0.00	+2.29

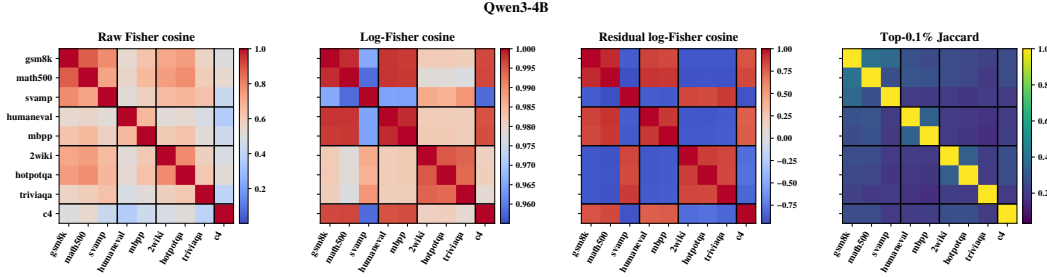


Figure 3: Capability regions in gradient space for Qwen3-4B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

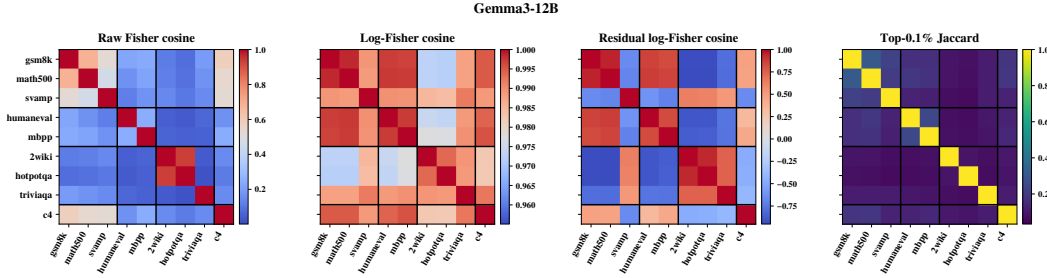


Figure 4: Capability regions in gradient space for Gemma3-12B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

Zeroing 0.05% of weights, selected from gradients alone, selectively removes one capability while leaving the others essentially untouched (the QA ablation moves code by +0.0001). Pending multi-model replication and the control ablations above, we describe the regions as causally implicated and partially separable rather than as proven causal carriers; if confirmed, ablation itself is a form of targeted scaling-down.

B.2 The pruning law

The mechanism form of §4.3 fits the entire pre-cliff grid. With the measured alignment term subtracted, the curvature residual follows a power law in the deleted capability mass $m_c(d)$:

$$\Delta L_c(d) = \underbrace{\bar{g}_c^\top \delta w_d}_{\text{measured}} + B \left[\underbrace{m_c(d)}_{\text{measured}} \right]^\gamma, \quad (7)$$

with in-sample $R^2 = 0.82\text{--}0.996$ over 24 capability $\times$ model fits. The held-out battery (1000-resample cell bootstrap) is where the form proves itself: in leave-one-family-out transfer the shared- γ mechanism form attains MAE 2.17 nats versus 10.3 for per-capability raw sparsity and 8.2 for the direct-damage baseline (both CIs exclude zero), so the deleted-mass coordinate is what transfers across families; it also beats the direct-damage baseline in fit-shallow/predict-deep (3.11 vs. 4.24) and raw sparsity in leave-largest-model-out (1.41 vs. 2.15), while its edge over raw sparsity within-model does not reach significance (without the first-order term, QA is unfittable, $R^2 \leq 0.67$; with it, QA joins the law). Two structural results: (i) in the deleted-mass coordinate the exponent concentrates at $\gamma \in [1.2, 2.2]$ across all families, versus 3.2–8.7 in the raw $(1-d)$ coordinate, which is what makes the form portable; and (ii) within a model, γ is shared across capabilities (e.g. Qwen3-0.6B: 1.41/1.56/1.43; Gemma3-4B: 1.66/1.71): capability differences are carried entirely by the two measured quantities, and the fitted structure is two parameters per model. Published

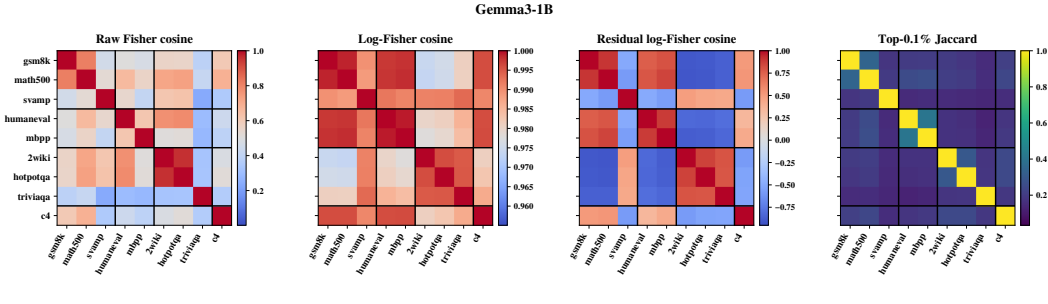


Figure 5: Capability regions in gradient space for Gemma3-1B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

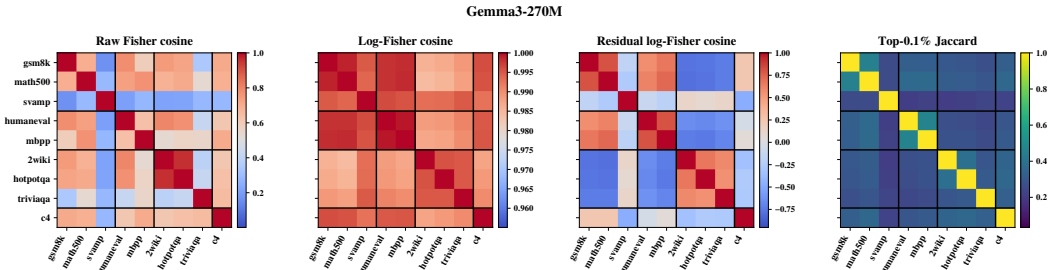


Figure 6: Capability regions in gradient space for Gemma3-270M: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

(reduced) forms substitute the smooth dependence of the geometry itself on (N_0, f) , leaving only basic parameters in the interface.

B.3 The signed first-order term predicts, in both directions

The alignment term $\bar{g}_c^\top \delta w$ is the decisive family coefficient. Its sign predicts the presence or absence of the capability-selective “free lunch” in every case measured: the QA term of Qwen3 is negative and QA improves under pruning (predicted -0.83 vs. measured -0.79 at 4B, $d = 0.7$); the QA term of Gemma is positive and no improvement exists at any size; the term for OLMo-3-7B is positive (mild degradation, prediction $+0.015$ = measurement at $d = 0.8$, code) while the term for OLMo-3-32B is negative, and deep-pruning QA improvement duly appears. At the level of individual model $\times$ capability $\times$ density cells in the mild regime, sign prediction accuracy is 67.6% (Wilson 95% CI [58.3, 75.7], $n=108$), above chance, with the family-level pattern (Qwen and OLMo-32B QA improvements, Gemma none, an OLMo size flip) called in advance. Beyond the mild regime, first- plus second-order predictions fail by orders of magnitude at the cliff in every family: the collapse is non-perturbative in loss space, confirming the two-regime structure of equation 7.

B.4 The cliff is a geometric threshold crossing

Measuring the capability geometry of the pruned models resolves the apparent unpredictability of the cliff. The capability-region mass retention (the fraction of the capability-exclusive top coordinates of the dense model still prominent after pruning) declines smoothly from the first densities (no cliff in geometry) while loss is still flat; the behavioral cliff occurs where retention crosses a critical value. The critical value is family-specific (Gemma: 0.035–0.10 across 1B/4B and all capabilities; Qwen: 0.13–0.23) but consistent within a family

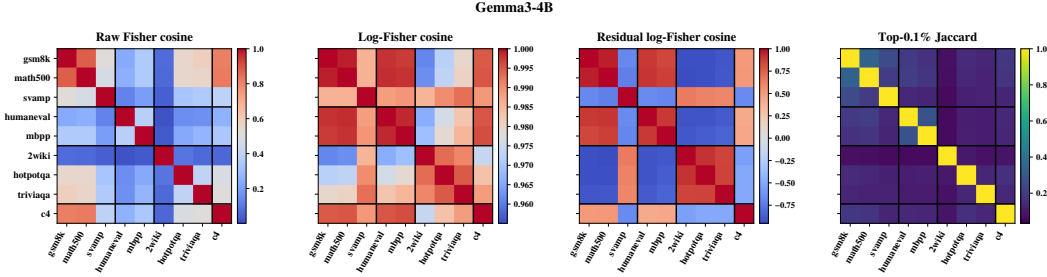


Figure 7: Capability regions in gradient space for Gemma3-4B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

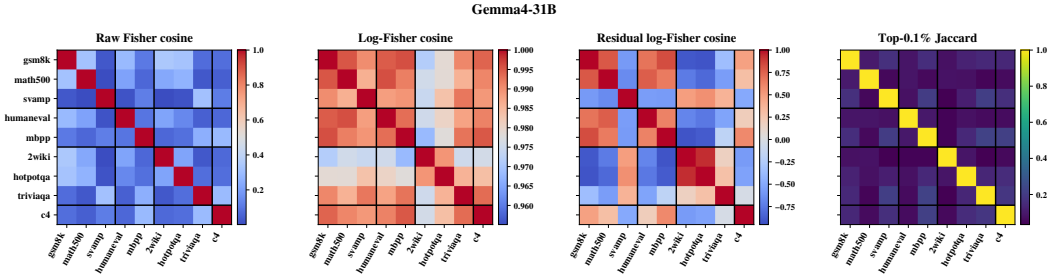


Figure 8: Capability regions in gradient space for Gemma4-31B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

across sizes and capabilities: the cliff boundary $d_c^*(N_0, f)$ reduces to one critical-retention constant per family, turning the least elegant fitted parameter of the two-regime law into a measurable, smoothly evolving order parameter. (preliminary: three models, two families; multi-model extension queued)

B.5 Recovery arm: redeemable damage and the asymptotic residual

Recovery training on general text fits the recovery law cleanly (Gemma3-270M pruned to $d = 0.6$; $R^2 \approx 1.0$): $\beta'_{\text{math}} = 1.23$, $\beta'_{\text{code}} = 1.02$, with asymptotic residuals under the fixed LoRA recipe, $L_c^\infty - L_c^{\text{dense}}$, of 1.21 (math) and 1.93 (code) nats; about 80% of this pruning damage is redeemed within 8M tokens and the remainder persists under this recipe, while QA recovers completely. (preliminary: one model, one density, one recipe; with three-point ladders the largest-budget extrapolation of the saturating form beats a carry-forward baseline in only 4 of 9 fits, so the ladder is too sparse to claim extrapolation, and the denser multi-model grid (B7) is queued) Two mechanism results: recovery on aligned data (teacher traces) heals substantially faster at small budgets (math 2.76 vs. 3.47 at 0.5M tokens), as the gradient-overlap prediction requires, but over-training on narrow aligned data re-damages the model (QA 4.95 $\rightarrow$ 9.75 from 0.5M to 8M tokens), while general-text recovery is monotone. Aligned recovery is fast but risky; general recovery is slow but safe; the optimal recovery budget is finite for narrow data. Quantization damage past the bit cliff is markedly less redeemable under the same recipe (residuals ≈ 2.6 –2.8 after int3) than pruning damage at matched severity.

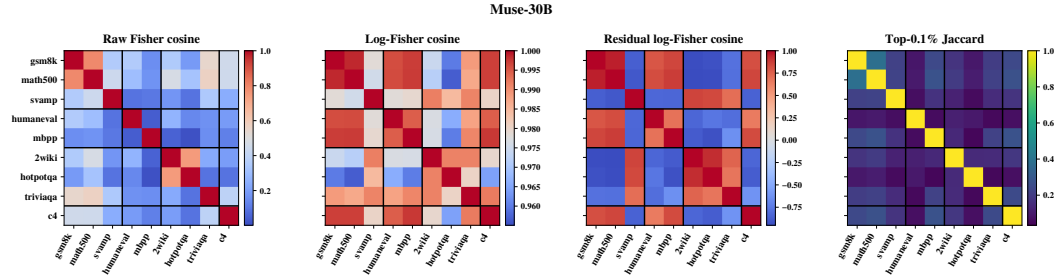


Figure 9: Capability regions in gradient space for Muse-30B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

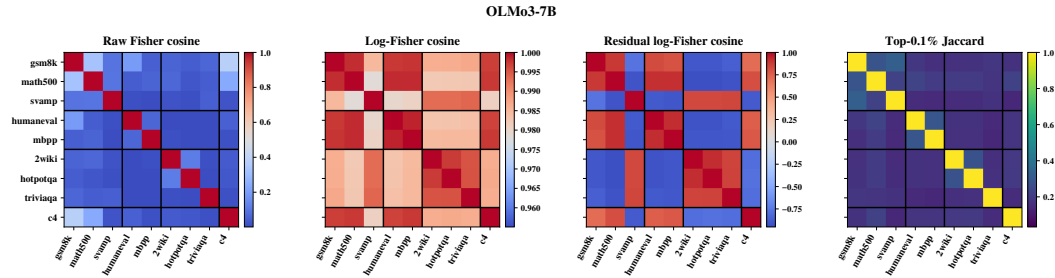


Figure 10: Capability regions in gradient space for OLMo3-7B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

B.6 Scale and family structure

Within-family robustness is non-monotonic in two families independently. Gemma-3 robustness peaks at 12B (27B collapses far earlier, +4.0 vs. +0.68 at $d = 0.7$), mirroring the prior ability-space observation of a Qwen3 14B peak. A mid-scale robustness peak is a reproducible pattern of scaling-down.

A candidate predictor, honestly retired. The top-concentration statistic κ_c tracked robustness within the Qwen3 ladder and was pre-registered as a predictor; the full grid retires it (cross-family inversion: OLMo, the most robust family, has the lowest concentration; within-Gemma top-end violation).

Matched-size panel ($\sim 30B$). At matched scale the family coefficient dominates: ΔL_{math} at $d = 0.7$ spans +0.02 (OLMo-3-32B), +0.17 (Muse-30B), +0.68 (Gemma4-31B), +4.04 (Gemma3-27B). OLMo-3-32B survives 70% pruning with $\leq +0.63$ nats on every capability. Family provenance (data recency, over-training exposure) is a first-order determinant of prunability, and “which family should an agent scale down from?” is a measured quantity.

B.7 Re-analysis of the prior ability-space grid

A prior internal study fit compression laws directly on Rasch abilities from ~ 100 -item panels over a 774-cell Qwen3 grid and reported that no derived law had held-out predictive power. Our re-analysis attributes much of this to the measurement layer: (1) the $r=1$ falsification anchor was driven entirely by the answer-only recipe (stratified, the anchor holds and held-out error drops from $9.0\times$ to $1.8\times$ the noise floor); (2) same-benchmark evaluation inflated QA gains by roughly one third; (3) discrimination heterogeneity voids interval-scale ability

1080 while rankings survive ($\rho \geq 0.98$); (4) item-level damage is smooth pre-cliff, ruling out
1081 discrete skill deletion at the item level. These findings fixed the protocol of §3.
1082
1083
1084
1085
1086
1087
1088
1089
1090
1091
1092
1093
1094
1095
1096
1097
1098
1099
1100
1101
1102
1103
1104
1105
1106
1107
1108
1109
1110
1111
1112
1113
1114
1115
1116
1117
1118
1119
1120
1121
1122
1123
1124
1125
1126
1127
1128
1129
1130
1131
1132
1133